Huỳnh Trần Anh Thư

DATA ANALYST

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PORTFOLIO | GITHUB

SUMMARY

Final-year Information Systems student with hands-on experience in collecting, cleaning and analyzing data, building dashboards and turning findings into business recommendations across logistics, retail and banking. Seeking an opportunity to apply and further expand my analytical skills to solve real-world business problems. In the next 2–3 years, I aim to grow into a Full-Stack Data Professional across analytics, engineering, BI, and applied AI.

EXPERIENCE

DATA ANALYST INTERN — SMARTLOG **May 2026 – Aug 2026**

Panasonic – Route to Customer | TMS Auto Planning Analytics – SQL, Metabase, Excel

- Owned analytics for the **TMS Auto Planning (route optimization) engine** serving Panasonic's distribution network: **12 carriers** across **17 delivery zones** and **~100 delivery trips daily**.
- **Built automated monitoring** in Metabase to measure how far planners' final plans deviate from the algorithm, defining KPIs for **plan accuracy, constraint violations, cost/CBM drift, carrier reassignment** and **planning lead time**.
- Found that only **68% of auto plans were kept**, with accuracy ranging from **27% to 98% by zone**, and that **27% of adjusted shipments still broke operating constraints** (route-sequence rule most often), pinpointing the zones and rules for **algorithm recalibration**.
- Tracked **system adoption** (daily active users, runs, **~88% run success rate**) and ran monthly **data accuracy checks** and **On-time SLA** analysis to flag operational bottlenecks.

PROJECTS

ITB CUSTOMER FINANCIAL HEALTH ANALYSIS (RUNNER-UP) **September 2026**

Consumer Finance Analytics – SQL, Python, Excel, Scikit-learn | GitHub

- **Goal:** assess the financial health of **999 Home Credit customers** from **1.85M card transactions** (VND 3,244.6B) and design a customer-care strategy; owned the work end to end, from SQL data prep and EDA to recommendations.
- **Key insight: spend-to-income alone explains 81%** of the health score (ρ = −0.95); "tough months" jump from near zero below 70% of income to **93% above 87%**, an **early-warning threshold**. The December peak comes from **2.9× more transactions**, not bigger baskets, and **online spend share** drives engagement most (ρ = +0.83).
- **Impact:** segmented customers with K-Means on 8 behavioral features (ARI 0.97 across 50 bootstraps) into a **5-segment support strategy** with **6 non-punitive interventions** in a 10-week rollout covering **90.9% of customers**.

VINDATATHON 2026 – E-COMMERCE ANALYTICS & FORECASTING (TOP 20) **2026**

Retail / Fashion E-commerce – Power BI, DAX, Python, Scikit-learn, LightGBM | GitHub

- Validated 13 tables (2.96M rows) with **automated data quality checks** and built a **5-page Power BI dashboard** with **46 DAX measures** on a star schema covering **10.5 years** of a fashion e-commerce platform's data.
- **Diagnosed the decline:** traffic grew 63% while **conversion fell from 1.01% to 0.29%** and margin from **22% to 14%**: a conversion and retention problem, not traffic.
- **Segmented customers** with K-Means on PCA of log-scaled RFM, choosing **k = 5** by silhouette, Davies-Bouldin and stability (**ARI 0.956**): **Champions and Loyal are 26% of customers but 78.5% of revenue**.
- Forecast 548 days of revenue with a **Ridge + LightGBM ensemble (R^2 0.807)**, then combined forecast and segments into a **7-step action plan** with tracking KPIs.

RESEARCH

PERSONALIZED FEDERATED HETEROGENEOUS GRAPH LEARNING FOR CREDIT RISK **Aug 2026 – Present**

Graduation Thesis – Python, PyTorch Geometric, LightGBM, Scikit-learn | GitHub

- **Problem:** lenders cannot pool customer data, so small lenders judge risk from too little data. Linked 7 Home Credit tables (**307,511 customers, 26.3M records**) into a **heterogeneous credit graph** (6 node, 10 edge types), keeping orphan records via placeholder nodes, plus **601 tabular features** for Logistic Regression and LightGBM baselines.
- Simulated **5 lenders** by region (3.2%–47.4% of customers) with an encoder fitted from aggregate statistics only, and trained a 2-layer **HeteroGNN** under Local-only, **FedAvg**, FedProx, FedPer and Ditto on one shared test set (paired bootstrap, 3 seeds).
- **Results:** FedAvg reached **0.7836 ROC-AUC**, on par with centralized training (0.7858), and beat local-only models at all 5 lenders (**up to +0.03 ROC-AUC** for small lenders), significant at **4 of 5**. Flagged applicants default at **27%, 3.4× the 8.1% base rate**, catching 45% of defaulters.

COMPETITIONS

• **ITB [UEL]** — Runner-up of 200 • **VinDatathon [VinUniversity]** — Top 20 of 500 • **Future Business Analyst [RMIT]** — Top 15

KEY COMPETENCIES

Technical: SQL (SQL Server, PostgreSQL; Window Functions, CTEs, Data Quality Checks), Python (Pandas, NumPy, Scikit-learn, LightGBM), Power BI (DAX, Data Modeling), SSIS/SSAS, Excel, Looker, Metabase, Kafka, PySpark, Git
Domain Knowledge: Banking & consumer finance (financial health, credit risk); Logistics & distribution (TMS, SLA, on-time delivery, vendor performance); Retail & e-commerce (conversion funnel, retention, RFM, AOV, returns)
Soft Skills: Stakeholder communication and KPI alignment, data storytelling, ownership of end-to-end analysis, structured problem solving, teamwork under tight deadlines, **self-learning with AI tools**

EDUCATION & CERTIFICATIONS

University of Information Technology – VNU-HCM Bachelor of Information Systems, 2023 – Present
GPA – 3.2 | **CFA Level I** – Candidate (in progress) **IELTS Academic** – Overall 6.5